# Bohdan Barulin

Backend Developer (Python / FastAPI)

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## Summary

Backend developer with hands-on experience in system architecture, high-load data collection pipelines, and business process automation. Built complete solutions solo from scratch: from database design to VPS deployment and custom hardware-software integrations. Specialized in Python (FastAPI), PostgreSQL, and resilient scraping architectures; also build tooling in Go. Focused on resource efficiency and building solutions that measurably save businesses time and money.

## Skills

**Languages & DB:** Python (FastAPI), Go, PostgreSQL, SQLite

**Architecture & Infrastructure:** Docker, WebSockets, Cloudflare R2, Hetzner VPS, background jobs (Procrastinate), process managers (temps), Let's Encrypt, Tailscale, Cloudflare Tunnel

**Scraping & Automation:** HTML/DOM parsing, Camoufox (anti-detect browser), browser fingerprint management, rotating proxy pools, scrcpy (Android control)

**Integrations:** RESTful APIs (payment gateways, logistics providers, SMS verification services, Marketing & Ads APIs), GitHub Apps, digital signatures (EDS)

**Frontend (Full Stack approach):** HTMX, Alpine.js, Tailwind CSS

**Languages:** Ukrainian (native), English (B1 — Intermediate)

## Experience

**Backend Developer**, Freelance / NDA Dec 2025 – Aug 2026

- **Architecture & development:** Designed and built the backend of an end-to-end lead generation and outreach automation system (FastAPI, PostgreSQL, HTMX, Alpine.js, Tailwind). Set up VPS deployment (Hetzner) with the temps process manager; contributed to the tool's open-source repository by reporting issues.
- **High-load scraping:** Built a block-resistant scraper for 3 major classifieds websites (HTML/DOM extraction). A single 2-hour run processes 60,000 listings from one source and identifies up to 10,000 new records. Implemented a pool of 250 rotating proxies with cooldown scheduling.
- **Data engineering:** Reverse-engineered the client-side data rendering of a major automotive marketplace to reliably extract structured listing data. Built and maintained a proprietary database of 450,000+ seller records with listing history and activity tracking.
- **Data processing optimization:** Introduced asynchronous task processing with the Procrastinate queue for fast contact filtering. Automated export of validated data in VCF format to Cloudflare R2.
- **Hardware-software integration:** Integrated physical Android smartphones via a USB hub using scrcpy and a WebSocket agent to automate messaging. Processing 1,000 contacts per day, previously done by 3 managers in 9 hours, is now supervised by 1 person in 3 hours.
- **Marketing integrations:** Set up automatic bi-weekly sync of collected contacts with Facebook Custom Audiences for precise ad targeting.

**Automation Developer**, Freelance Spring 2026

- Developed a browser automation pipeline for multi-step user flows (sign-up, verification, data collection) on a platform with strict anti-bot protection.
- Used Camoufox with browser fingerprint management to keep sessions stable; ran 10 parallel workers handling up to 100 flows per day.
- Integrated third-party SMS verification APIs to fully automate OTP handling.

## Projects

**hakobu — self-hosted deployment tool (Go, open source, MIT)** 2026 – present

- Deploys apps from GitHub to servers you own: on a push to the default branch it builds the new version and swaps it in with no downtime, behind a Cloudflare Tunnel, so the server needs no open ports. Site and docs: hakobu.dev.
- Blue/green deploys with rollback, including the database.
- Postgres, Valkey and R2 storage provisioned per project; encrypted backups that are test-restored after every upload.
- Sentry-compatible errors, logs and traces kept on your own server.
- Ships as one binary with SQLite state, runs apps in rootless Docker, and publishes signed releases.

## Education

**Lutsk National Technical University (LNTU)**, BSc in Computer Science 2024 – 2028

- **2nd place, Django Pub Hackathon:** In a 3-person team, built and presented in 3 days a working user authentication module based on qualified electronic digital signatures (EDS).